PROJECT AND TEAM INFORMATION

Project Title

InternMatch: A Real-Time Internship Recommendation System using DSA and OOP

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

- *While searching for internship opportunities, students have to go through large and unstructured databases and manually check the eligibility criteria of each internship, which is time-consuming and inefficient, often leaving them unsure which internship best fits their skill set.*
- *InternMatch resolves this by automatically verifying eligibility and calculating a match percentage based on skill matching, while also listing missing skills the student can work on.*

State of the Art / Current solution

- *Currently, students refer to platforms like LinkedIn, which do not check individual student eligibility or provide a skill match percentage.*
- *Our project instead utilises a combination of DSA and OOP concepts with a simple rule-based approach, fetching live-time data from public job listings at runtime to verify eligibility, find the match percentage, and identify the skill gap.*

Project Goals and Milestones

Project Goal: *The goal of InternMatch is to create a system that fetches real-time internship data using an API, takes student details dynamically (CGPA, branch, skills), checks eligibility, calculates the match percentage, and shows the best matching internship opportunities.*

Phase 1 (Ongoing):

- *Finalize requirements and workflow; design the Student and Internship classes.*
- *Implement real-time internship data fetching using a public API and handle it using JSON.*

Phase 2:

- *Store internship data in a Linked List; check eligibility based on CGPA and branch.*
- *Implement skill searching/matching, calculate match percentage and skill gap, and implement sorting.*

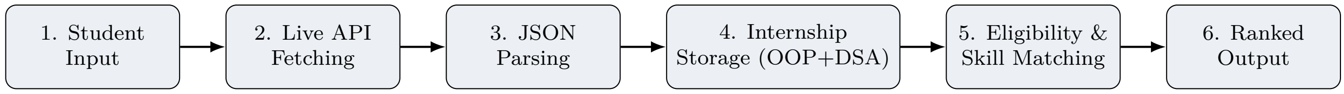
Phase 3:

- *Recommend internships based on match percentage and show the skill gap as final output.*
- *Test with multiple student profiles; upload the project on GitHub with complete documentation.*

Project Approach

- *Our project will be developed as a C++ console-based application using OOP and DSA concepts, with separate Student and Internship classes to store and manage information.*
- *Instead of pre-stored data, the system fetches live internship listings from the Arbeitnow API using libcurl, processed with nlohmann/json, and stores them in a linked list.*
- *The system then checks eligibility based on CGPA and branch, compares skills using vectors and unordered sets, and sorts internships by match percentage.*
- *We will use VS Code for development and GitHub to maintain the code and documentation.*

System Architecture (High Level Diagram)



The system takes student input (CGPA, branch, skills), fetches live internship data via the Arbeitnow API using libcurl, parses it with nlohmann/json, stores records using a linked list (OOP+DSA), checks eligibility and matches skills, and presents a ranked list of best-matching internships.

Project Outcome / Deliverables

- *Our project is a C++ console-based application that takes student details (name, branch, CGPA, skills) and fetches live internship listings instead of pre-stored data.*
- *It checks eligibility based on CGPA and branch, compares skills to show the match percentage and missing skills, and provides a ranked list of suitable internships.*
- *The project's source code and documentation will be maintained on GitHub.*

Assumptions

- *We assume the API will be accessible and working properly, providing sufficient internship information, and that the user has a stable internet connection.*
- *Since the API may not always provide specific details like CGPA or branch eligibility, we use the available skill tags and internship requirements as a reasonable way to determine match quality.*

References

1. *libcurl Documentation – https://curl.se/libcurl/*
2. *nlohmann/json Library – https://github.com/nlohmann/json*
3. *Arbeitnow Job Board API – https://www.arbeitnow.com/api/job-board-api*